

Introduction and Highlights of ECC and ECC-PLC



Function Introduction of ECC

The gateway ECC is an energy control center which collects PV production data and energy consumption data and delivers it to the SolaX Cloud via flexible networking with Wi-Fi, 4G and Ethernet. ECC enables users to conveniently manage remote operation and maintenance of their PV system anytime and from anywhere by SolaX Cloud or SolaX App.

Function Introduction of ECC-PLC

SolaX gateway ECC-PLC is used to collect PV production data and energy consumption data and delivers it to the ECC via RS485. Besides, it support bidirectional communications for remote maintenance. With integrated optional production metering and consumption monitoring, the ECC-PLC is the platform for smart energy management.

Highlights and key Features of ECC

- Load consumption monitoring
- Flexible networking with Wi-Fi, 4G and Ethernet
- 5V USB port available
- Internal relay available to control external devices
- Support seamless communication with peripherals through RS485 and Ethernet
- Support remote operation and maintenance

Key Features of ECC

1. Monitoring and Management
 - The system capacity: 40 microinverters with built-in PLC.
 - Support local/remote setting and monitoring.
 - Support batch upgrade.
2. Export Control
 - Limit power that exported to the electricity grid by adjusting the production power of the inverters.
 - The system requires ECC,ECC/PLC,A1-Micro 1 in 1*.
 - Fast responds to avoid the unexpected feed-in especially when the loads drop rapidly.

Export Control for WiFi Microinverter

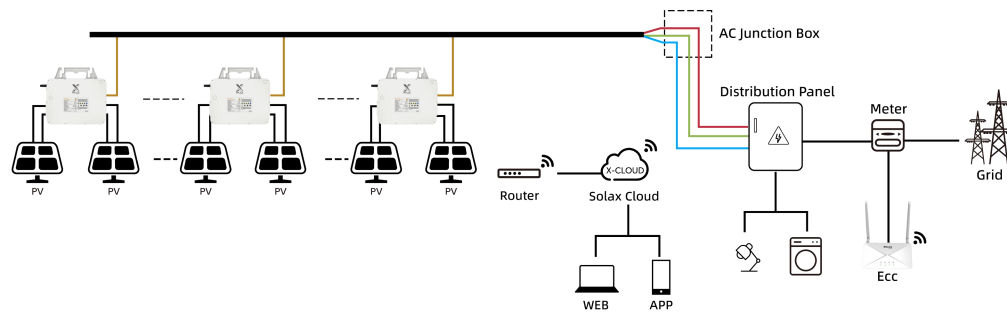


Figure 1 Export Control for WiFi Microinverter

Export Control for PLC Microinverter

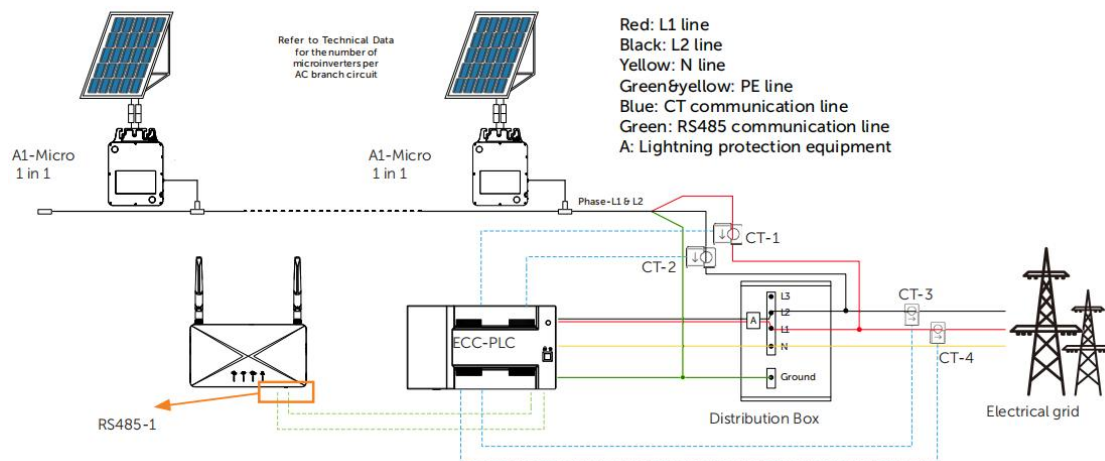


Figure 2 Export Control for PLC Microinverter

3. Ripple Control (being developed)

- The Ripple Control Receiver is required to implement the ripple control function.
- Ripple Control Receiver is a receiving device that can analyze the high-frequency signal sent from grid operator and trigger different status of switches accordingly.
- Each status can be set and assigned to a power control action.
- In total 16 status and actions can be programmable.

4. Local Data Storage

- Support up to 64GB TF card to store the data of 40 A1-Micro 1 in 1 units for 90 days
- Ideal for the offline site management

5. Microinverter Consumption Monitoring (without PLC)

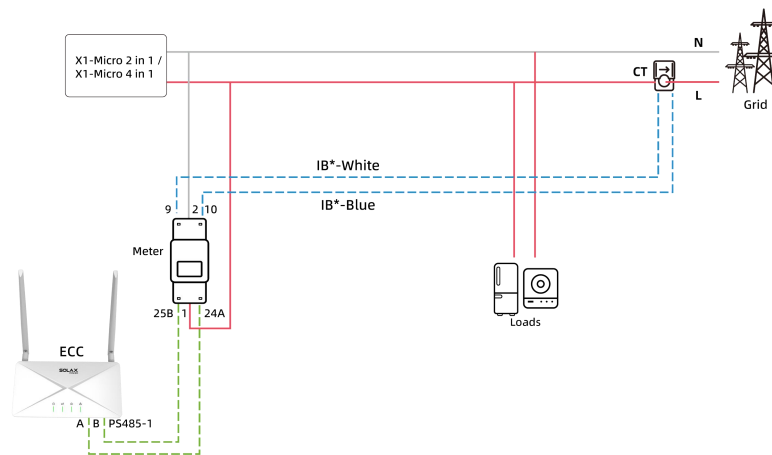


Figure 3 Microinverter Consumption Monitoring (without PLC)

- Currently compatible inverters :X1-Micro 2 in 1,X1-Micro 4 in 1
- Support export ripple control/DRM
- The ECC and Microinverter will automatically check the time with web after powering on.and the data of meter and microinverters will be independently to the solax cloud every 5 minutes in integer time.

6. Microinverter Consumption Monitoring (with PLC)

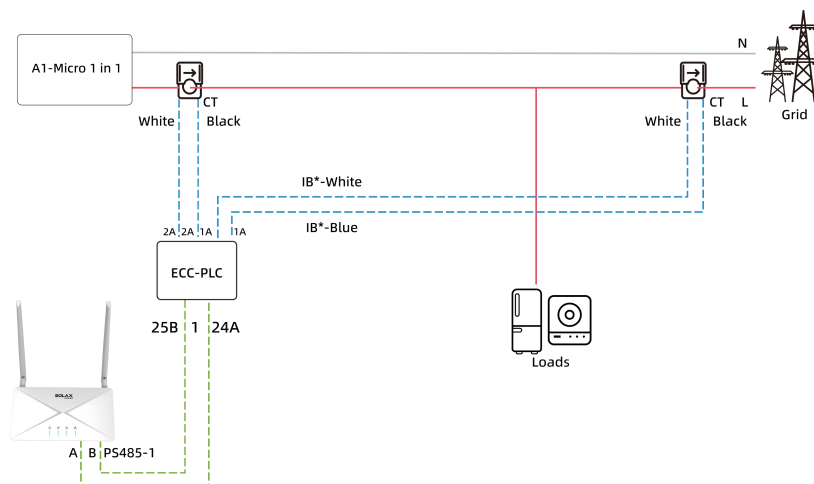


Figure 4 Microinverter Consumption Monitoring(with PLC)

- Currently compatible inverters: A1-Micro
- Support export control/ripple control/DRM
- By connecting Production CT and Consumption CT on the grid-connected side and PV side, ECC-PLC can transmit the collected microinverter power

generation and grid-connected power data together with the microinverter data to ECC and then to the solax cloud.

Highlights of ECC-PLC

- Real-time load and PV production monitoring
- Web-based monitoring and control
- Bidirectional communication for remote upgrades
- Built-in industrial-grade PLC module
- Support for remote operation and maintenance

Interface Layout of ECC

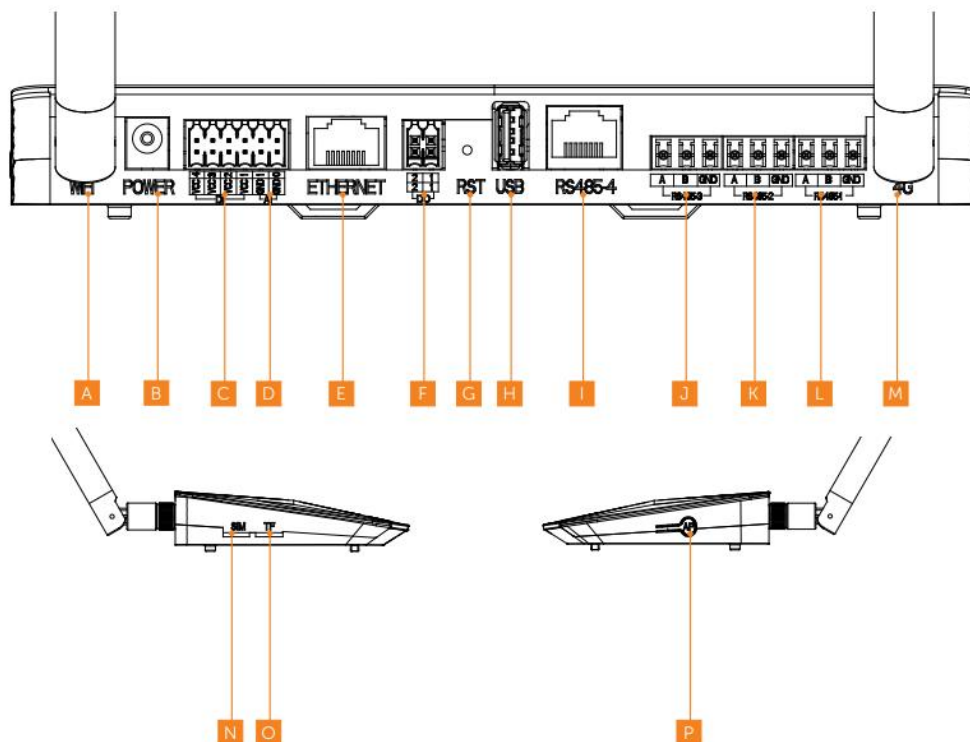


Figure 5 ECC Interface Layout

Table 1 ECC Interface Layout

Item	Description
A	Wi-Fi antenna
B	12V power input

C	DI socket(DI)
D	AI socket(AI)
E	NET socket(NET)
f	DO socket(DO)
G	RST button(RST)
H	USB socket(USB)
I	RS485 port-4
J	RS485 port-3
K	RS485 port-2
L	RS485 port-1
M	4G antenna
N	SIM card slot(SIM)
O	TF card slot(TF)
P	AP button

Note:

There are 4 strings of DI ports for DRM including DRM0/5/6/7/8 which responds according to grid-connection needs

There are 2 strings of AI ports

There are 2 strings of active DO ports (12 V) used for dry contact and adapter box

RS 485 port-4 is used with adapter box for communication

RS 485 port-3 is used with AC-Coupling communication with energy storage system

RS 485 pORT-2 is used with for modbus communication with the third party host (to measure wind speed, temperature and radiation strength).

RS485 port-1 is used with wired meters and to communicate with ECC-PLC and wired meter.

Interface Layout of ECC-PLC

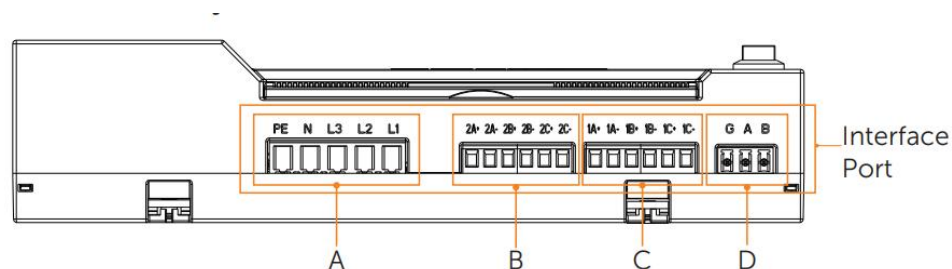


Figure 6 ECC-PLC Interface Layout

Table 2 ECC-PLC Interface Layout

No.	Item	Description
A	Power	For power supply connection
B	Current sampling port	For production CT connection
C	Current sampling port	For consumption CT connection
D	RS485 port	For RS485 connection